

Trajectory-Space Omega Triage: A Branch-Selection Research Note

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Abstract

The current Omega program has isolated a quotient-relative COM fiber-transport witness in a toy multifield substrate. A natural parallel branch asks whether Omega can be formulated more directly in trajectory space, reducing dependence on hand-chosen macro-quotients. The most elegant trajectory-space candidate is conditional temporal dependence under viability, but direct conditional mutual information over trajectory segments is likely estimator-sensitive and sample-hungry.

This note proposes a small branch-selection probe rather than an immediate commitment to any single trajectory-space formalism. The probe compares four candidate readout families: viability kernel/hazard profiles, trajectory concentration and collapse geometry, recoverability or viable tube thickness, and predictive temporal dependence surrogates. The goal is to identify which trajectory-space route has enough signal, stability, interpretability, and computational practicality to justify deeper work. The probe is designed to be minimal, principled, and revelatory: it may support a trajectory-space pivot, refine the COM witness, or show that the quotient/fiber route remains the best operational branch for now.

Status and Scope

This note documents a proposed triage probe for the trajectory-space branch of Omega. It does not replace the current COM fiber-transport artifact. The COM witness remains the best-audited operational signature. The purpose of this branch is to test whether the COM witness is an operational discretization of a deeper trajectory-space object, or whether the trajectory-space route reveals a different or more fundamental formulation.

The motivation for this branch comes from the structural quotient problem: the current fiber construction measures viable propagation only after choosing a macro-map. A trajectory-space formulation may reduce that dependence by working directly with viable trajectory measures, hazard profiles, recoverability neighborhoods, concentration geometry, and temporal prediction.

However, trajectory-space methods introduce their own difficulties. Continuous trajectory measures require choices of metric, density estimator, coordinate representation, and finite-sample approximation. The triage probe is therefore not designed to prove a trajectory-space Omega object. It is designed to choose which trajectory-space direction deserves deeper empirical investment first.

1 Motivation

The current Omega witness is quotient-relative:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa\text{COM}}(S_{\text{toy}}). \quad (1)$$

This is a real operational achievement, but it leaves the quotient problem unresolved. The long-term target remains a universal or quotient-stable signature of viable propagation. A trajectory-space branch tests whether the object can be approached before choosing a macro-fiber quotient.

A first idealized trajectory-space candidate is:

$$I_t^V = I(\tau_{0:t}; \tau_{t:T} \mid V(\tau_{0:T}) = 1), \quad (2)$$

which measures how much the early portion of a viable trajectory informs the later portion, conditioned on full-horizon viability.

This is elegant, but difficult. Conditional mutual information over trajectory segments is sample-hungry, estimator-sensitive, and vulnerable to implicit representation choices. A Gaussian approximation may fail in the non-Gaussian regimes of greatest interest, while binning continuous trajectories reintroduces a quotient-like choice. Conditioning on entire partner trajectories in the multifield case may also remove coupled structure rather than detect component preservation.

The triage probe therefore treats full conditional mutual information as a long-term target rather than the first empirical backbone. It asks which simpler trajectory-space readout should be developed first.

2 Candidate Trajectory-Space Profile

The trajectory-space branch should initially use a profile rather than a single scalar:

$$\Omega_{\text{traj}}(s, T) = (p_T, h_T, Q_T, R_T, A_{\text{pred}}, C_{AB}, K_T). \quad (3)$$

The coordinates are:

- p_T : viable trajectory mass;
- h_T : survival or first-failure hazard profile;
- Q_T : trajectory concentration or collapse geometry;
- R_T : recoverability or viable tube thickness;
- A_{pred} : predictive temporal dependence surrogate;
- C_{AB} : multifield component dependence or component balance;
- K_T : comparison to product, shuffled, or time-shuffled null baselines.

The profile form is deliberate. A regime can survive while collapsing, spread while losing recoverability, predict while becoming deterministic, or preserve a joint aggregate while erasing one component. Different failure modes appear in different coordinates.

3 Probe T0: Trajectory-Space Branch Triage

3.1 Purpose

Probe T0 is a branch-selection probe. It compares the most promising trajectory-space readout families on a small set of known diagnostic conditions.

The purpose is not to validate a final trajectory-space Omega object. The purpose is to decide which trajectory-space path has enough signal, stability, interpretability, non-redundancy, and computational practicality to deserve deeper work first.

The candidate readout families are:

1. viability kernel, survival, hazard, and kernel erosion;
2. trajectory concentration and collapse geometry;
3. viable tube thickness;
4. restoration or return-to-corridor dynamics;
5. predictive temporal dependence surrogate.

The probe should answer:

Which trajectory-space route best separates robust viable propagation from survival-only, trap, noise, collapse, and COM-positive regimes?

3.2 Why Full Conditional Mutual Information Is Deferred

The idealized information-theoretic target remains:

$$I_t^V = I(\tau_{0:t}; \tau_{t:T} \mid V(\tau_{0:T}) = 1). \quad (4)$$

However, direct conditional mutual information over trajectory segments is likely to be sample-hungry and estimator-sensitive. Gaussian approximations may fail in non-Gaussian viability-boundary regimes, while discretized estimators reintroduce binning choices similar to quotient choices.

Probe T0 therefore treats conditional mutual information as a long-term target. The first empirical step uses more stable trajectory-space readouts: viability/hazard, kernel erosion, concentration, tube thickness, restoration, and predictive loss advantage.

3.3 Compute Budget

The probe should be small enough to run quickly but large enough to estimate stability:

$$\text{soft limit} = 3600 \text{ seconds}, \quad (5)$$

$$\text{hard limit} = 7200 \text{ seconds}, \quad (6)$$

$$\text{workers} = 18, \quad (7)$$

$$N_{\text{traj}} = 5000, \quad (8)$$

$$N_{\text{seeds}} = 80, \quad (9)$$

$$N_{\text{bootstrap}} = 200. \quad (10)$$

Fallback settings:

$$N_{\text{traj}} = 3000, \tag{11}$$

$$N_{\text{seeds}} = 60, \tag{12}$$

$$N_{\text{bootstrap}} = 100. \tag{13}$$

If runtime is cheap, increase seeds before adding conditions.

4 Conditions

4.1 Single-Field Diagnostic Worlds

Use four known diagnostic worlds:

- `open_field`;
- `sink_trap`;
- `rigid_attractor`;
- `noise_swamp`.

Their intended roles are:

- `open_field`: broad viable baseline;
- `sink_trap`: viability collapse;
- `rigid_attractor`: survival with path collapse;
- `noise_swamp`: entropy or noise without coherent propagation.

4.2 Multifield COM-Positive Corridor

Use the known COM witness regime:

- F, T attractive corridor;
- $\alpha \in \{0.50, 0.525\}$;
- $T \in \{900, 1500\}$.

Baselines:

- `coupled`;
- `product`;
- `shuffled`;
- `time-shuffled`.

Optional additions if runtime is very cheap:

- $\alpha = 0.45$;
- $T = 2400$.

These should not be added if they interfere with the triage purpose.

5 Candidate Readout 1: Viability Kernel, Hazard, and Erosion

5.1 Question

Does viable trajectory mass and failure-time geometry distinguish open, trapped, noisy, rigid, and COM-positive regimes?

5.2 Viable Trajectory Mass

Let:

$$\mathcal{F}_T(s) = \{\tau_{0:T} : \tau_0 = s, V(\tau_{0:T}) = 1\}. \quad (14)$$

The viable trajectory mass is:

$$p_T(s) = \mu_s(\mathcal{F}_T(s)). \quad (15)$$

5.3 Survival and Hazard

The survival curve is:

$$S(t) = P(\tau_{0:t} \text{ is viable}). \quad (16)$$

The first-failure hazard is:

$$h(t) = P(\text{first failure at } t \mid \text{survived to } t - 1). \quad (17)$$

Metrics:

- p_T ;
- survival curve $S(t)$;
- first-failure hazard $h(t)$;
- tail survival;
- hazard peak;
- hazard AUC;
- late-horizon hazard slope.

5.4 Kernel Erosion Along Viable Trajectories

Survival asks whether a trajectory has remained viable so far. Kernel erosion asks how much viable future remains along a surviving trajectory.

For a viable trajectory $\tau_{0:T}$, define the remaining viable mass at time t as:

$$k_{\text{rem}}(t) = \mu_{s_t}(\mathcal{F}_{T-t}(s_t)). \quad (18)$$

Operationally, estimate $k_{\text{rem}}(t)$ by launching a small number of continuation rollouts from segment-boundary states s_t and measuring viability to the remaining horizon $T - t$.

Kernel erosion metrics:

- mean remaining viable mass by segment;
- minimum remaining viable mass along viable trajectories;
- kernel erosion slope;
- kernel erosion AUC;
- coupled-vs-null erosion delta.

Kernel erosion is the trajectory-space analogue of asking whether viable structure is carried forward rather than merely surviving to the next step.

5.5 Expected Separation

- Trap regimes should show a strong hazard peak or rapid survival collapse.
- Open regimes should show high survival and low erosion.
- Rigid attractors may show high survival but must be flagged by concentration metrics.
- Noise swamps may show unstable hazard or weak structured persistence.
- COM-positive regimes may show coupled survival, hazard, or erosion advantages relative to product and shuffled baselines.

5.6 Decision Value

This branch is promising if hazard and kernel erosion separate regimes cleanly and add information beyond terminal viability.

6 Candidate Readout 2: Trajectory Concentration and Collapse Geometry

6.1 Question

Can trajectory-space geometry distinguish robust viable corridors from survival by collapse into a narrow path?

6.2 Metrics

Use viable trajectories only. Compute:

- mean pairwise trajectory distance;
- median pairwise trajectory distance;
- endpoint variance;
- path covariance trace;
- effective dimension;

- concentration ratio.

Represent each trajectory as a vector in trajectory-coordinate space. Let λ_i be the eigenvalues of the covariance matrix of viable trajectory vectors. Define effective dimension:

$$d_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (19)$$

A simple normalized collapse diagnostic is:

$$Q_{\text{collapse}} = \frac{d_{\text{eff}}}{d_{\text{ambient}}}. \quad (20)$$

6.3 Expected Separation

- Rigid attractors should show high survival but low pairwise distance and low effective dimension.
- Noise swamps may show high spread without recoverability or prediction.
- Open fields should show broader viable spread.
- COM-positive regimes need not maximize spread, but should avoid total collapse into a single path.

6.4 Decision Value

This branch is promising if it cleanly separates survival-with-collapse from robust viable corridors.

7 Candidate Readout 3: Viable Tube Thickness

7.1 Question

Are viable trajectories embedded in neighborhoods of nearby viable continuations, or are they brittle single threads?

7.2 Conceptual Definition

Viable tube thickness asks whether a small perturbation of a viable trajectory remains inside a region with viable continuations. It measures local fatness of the viable set.

This should be distinguished from restoration. A system can have a fat viable tube without a restoring dynamic. Tube thickness measures robustness to local deviation, not return toward the original corridor.

7.3 Operational Version

For a sampled viable trajectory τ , perturb states at segment boundaries:

$$s'_{t_i} = s_{t_i} + \epsilon. \quad (21)$$

Then estimate whether a viable continuation remains available after the perturbation. Use perturbation radii:

$$\epsilon \in \{0.01, 0.03, 0.05\}. \quad (22)$$

For each viable trajectory and segment boundary, run a small number of continuation rollouts:

$$N_{\text{perturb_rollouts}} = 5. \quad (23)$$

7.4 Metrics

- perturbation survival rate;
- viable continuation probability after perturbation;
- viable tube thickness by segment;
- minimum tube thickness;
- tube thickness AUC over segment boundaries.

7.5 Expected Separation

- Trap regimes should have low tube thickness near sink boundaries.
- Rigid attractors may survive but have low local viable thickness.
- Noise swamps may show unstable or noisy thickness estimates.
- COM-positive regimes should show viable transport with nontrivial local viable neighborhoods.

7.6 Decision Value

This branch is promising if it separates robust viable corridors from brittle survival more clearly than survival or hazard alone.

8 Candidate Readout 4: Restoration and Return-to-Corridor

8.1 Question

If a viable trajectory is perturbed, does the dynamic return toward the original viable corridor or an equivalent viable trajectory family?

8.2 Conceptual Definition

Restoration asks whether a perturbation is actively corrected or reabsorbed by the dynamics. This differs from tube thickness. Tube thickness asks whether viability remains possible after deviation; restoration asks whether the system returns toward a coherent viable channel.

Omega-like propagation may require both:

- local thickness, so perturbations do not immediately destroy viability;
- restoration, so deviations do not drift into unrelated or brittle survival regimes.

8.3 Operational Version

For each perturbed rollout τ' , compare the perturbed continuation to the original viable trajectory family or reference corridor $\mathcal{C}$.

Let $d(\tau', \mathcal{C})$ denote distance from the perturbed continuation to the reference viable ensemble. A simple restoration score is:

$$R_{\text{restore}} = 1 - \frac{d_{\text{late}}(\tau', \mathcal{C})}{d_{\text{early}}(\tau', \mathcal{C}) + \epsilon}. \quad (24)$$

Positive values indicate return toward the viable corridor. Values near zero indicate no restoration. Negative values indicate drift away.

The reference corridor $\mathcal{C}$ can initially be represented by the viable trajectory ensemble for the same condition, using nearest-neighbor distance, local PCA distance, or another simple trajectory-distance approximation.

8.4 Metrics

- restoration ratio;
- return-to-corridor probability;
- late distance to viable ensemble;
- restoration half-life;
- restoration failure rate.

8.5 Expected Separation

- Trap regimes should show weak restoration after boundary perturbations.
- Rigid attractors may show strong return but should be checked against collapse geometry.
- Noise swamps may show poor restoration despite local spread.
- COM-positive regimes should show restoration without total collapse.

8.6 Decision Value

This branch is promising if it detects return-to-corridor structure that is not captured by viable tube thickness alone.

9 Candidate Readout 5: Predictive Temporal Dependence Surrogate

9.1 Question

Does past viable trajectory structure predict future viable trajectory structure better than null baselines?

9.2 Motivation

Full conditional mutual information,

$$I(\tau_{0:t}; \tau_{t:T} \mid V(\tau_{0:T}) = 1), \quad (25)$$

is the elegant target but may be too difficult to estimate reliably in the first pass. A predictive-loss surrogate is more stable.

9.3 Setup

Split trajectories at segment boundaries:

$$\tau_{0:t_i} \longrightarrow \tau_{t_i:T}. \quad (26)$$

Train simple predictors on viable trajectories. Use interpretable models first:

- linear regression;
- ridge regression;
- nearest-centroid or lightweight nearest-neighbor diagnostics, if cheap.

Predict future summaries rather than the full future trajectory initially:

- future endpoint;
- future segment means;
- future viability suffix length;
- future displacement;
- future path PCA coefficients.

9.4 Metrics

- future prediction MSE;
- null prediction MSE;
- prediction advantage versus time-shuffled baseline;
- prediction advantage versus product baseline;
- prediction advantage versus shuffled baseline;
- late-horizon prediction advantage.

Define prediction advantage:

$$A_{\text{pred}} = \frac{L_{\text{null}} - L_{\text{real}}}{L_{\text{null}} + \epsilon}. \quad (27)$$

9.5 Baselines

- time-shuffled past/future pairs;
- product baseline;
- shuffled joint baseline;
- constant mean predictor.

9.6 Expected Separation

- Real viable past should predict viable future better than nulls in coherent propagation regimes.
- Noise swamps should show weak prediction advantage.
- Rigid attractors may predict well but should be flagged by concentration collapse.
- COM-positive regimes should show prediction advantage plus noncollapse.

9.7 Decision Value

This branch is promising if it gives stable, interpretable temporal-dependence signal without requiring full mutual information estimation.

10 Non-Redundancy and COM Alignment

10.1 Non-Redundancy Measurement

Non-redundancy is scored using correlations between readout families across conditions. A readout is less valuable if it is nearly identical to terminal viability mass p_T .

The probe should compute:

$$\rho(X, p_T) \tag{28}$$

for each branch readout X , and rank correlations where sample size permits.

Required correlations include:

- p_T vs hazard AUC;
- p_T vs kernel erosion AUC;
- p_T vs effective dimension;
- p_T vs tube thickness;
- p_T vs restoration score;
- p_T vs prediction advantage.

A readout receives a stronger non-redundancy score when it:

- distinguishes regimes that have similar p_T ;
- has moderate or low correlation with p_T ;
- explains COM-positive behavior not captured by p_T alone.

10.2 Metric-Specific COM Alignment

COM comparison should be metric-specific rather than binary. The probe should compute correlations or rank alignments between the COM propagation index and each trajectory-space readout.

Required COM alignment entries include:

- COM index vs p_T ;
- COM index vs hazard AUC;
- COM index vs kernel erosion AUC;
- COM index vs effective dimension;
- COM index vs tube thickness;
- COM index vs restoration score;
- COM index vs prediction advantage.

This allows several distinct outcomes:

- COM aligns with kernel erosion, suggesting a viability-kernel backbone.
- COM aligns with tube thickness or restoration, suggesting recoverability is central.
- COM aligns with prediction advantage, suggesting temporal information flow.
- COM aligns with concentration, suggesting noncollapse geometry.
- COM fails to align with trajectory readouts, suggesting quotient-specific structure.

11 Branch Scoring

The purpose of Probe T0 is to choose a branch, not to maximize a single metric. Each readout family receives a score from 0 to 3 on five criteria:

1. separation;
2. stability;
3. interpretability;
4. non-redundancy;
5. compute practicality.

11.1 Score Scale

- 0: useless, noisy, or uninterpretable;
- 1: weak signal;
- 2: useful signal;
- 3: strong signal worth deeper probe.

11.2 Separation

Does the readout distinguish known regimes?

- open versus trap;
- rigid survival versus robust corridor;
- noise versus coherent propagation;
- COM-positive coupled condition versus product/shuffled nulls.

11.3 Stability

Are bootstrap intervals and seed variability tolerable?

- confidence intervals do not wildly cross the effect direction;
- seed behavior is stable;
- estimator warnings are not dominant.

11.4 Interpretability

Does the metric tell a clear story?

- hazard collapse;
- kernel erosion;
- path collapse;
- tube thickness;
- restoration;
- temporal prediction advantage.

11.5 Non-Redundancy

Does the metric add information beyond p_{viable} ? A metric that only tracks survival is less valuable for Omega.

11.6 Compute Practicality

Can the readout be scaled without estimator instability or excessive runtime?

12 Required Outputs

The result folder should be:

```
probe_T0_trajectory_space_branch_triage_results/
```

Required files:

- `summary.json`
- `branch_scores.csv`
- `viability_hazard_profile.csv`
- `kernel_erosion_profile.csv`
- `trajectory_concentration.csv`
- `tube_thickness_profile.csv`
- `restoration_profile.csv`
- `predictive_temporal_dependence.csv`
- `readout_correlations.csv`
- `com_metric_alignment.csv`
- `baseline_comparisons.csv`
- `bootstrap_intervals.csv`
- `estimator_report.csv`

Recommended plots:

- `survival_hazard_by_world.png`
- `kernel_erosion_by_world.png`
- `trajectory_concentration_by_world.png`
- `tube_thickness_by_world.png`
- `restoration_by_world.png`
- `prediction_advantage_by_world.png`
- `readout_correlation_matrix.png`
- `com_metric_alignment.png`
- `branch_score_radar.png`

13 Console Summary

The script should end with:

PROBE TO: TRAJECTORY-SPACE BRANCH TRIAGE

Runtime:

- status:
- workers:
- N_TRAJ:
- seeds:
- bootstrap repeats:

Branch scores:

- kernel/hazard/erosion:
- concentration/collapse:
- tube thickness:
- restoration:
- predictive temporal dependence:

Non-redundancy:

- readouts most correlated with p_viable:
- readouts least correlated with p_viable:
- readouts that separate similar-p_viable regimes:

COM metric alignment:

- strongest COM-aligned trajectory readout:
- weakest / conflicting readout:
- interpretation:

Best branch:

- name:
- why:
- strongest metric:
- weakest metric:
- estimator warnings:

Recommendation:

- deepen kernel erosion
- deepen concentration
- deepen tube thickness
- deepen restoration
- deepen predictive temporal dependence
- pause trajectory branch

14 Expected Outcomes and Decisions

14.1 Outcome A: Tube Thickness Wins

Proceed to:

Probe T1: Viable Tube Geometry

Focus on viable neighborhood thickness, perturbation survival, segment-wise tube erosion, and boundary proximity.

14.2 Outcome B: Restoration Wins

Proceed to:

Probe T1: Restoration and Return-to-Corridor Dynamics

Focus on restoration ratio, return-to-corridor probability, restoration half-life, and distinction from mere tube thickness.

14.3 Outcome C: Kernel Erosion Wins

Proceed to:

Probe T1: Viability Kernel Erosion Along Trajectories

Focus on remaining viable mass, erosion slopes, tail viability, and coupled-vs-null kernel deformation.

14.4 Outcome D: Concentration / Collapse Wins

Proceed to:

Probe T1: Viable Trajectory Geometry

Focus on effective dimension, pairwise trajectory distances, covariance spectrum, and collapse versus robust spread.

14.5 Outcome E: Predictive Temporal Dependence Wins

Proceed to:

Probe T1: Predictive Propagation in Trajectory Space

Focus on past-to-future prediction, time-shuffled nulls, long-horizon prediction, and component prediction balance.

14.6 Outcome F: No Trajectory-Space Readout Works

Pause trajectory-space scaling.

Interpretation:

The current simulator or data volume may not support trajectory-space inference yet.
The COM fiber witness remains the main operational object.

15 Methodological Guardrails

The triage probe preserves the three project principles.

15.1 Minimal

Only the most promising branch candidates are tested, using a small condition set. Full conditional mutual information is deferred.

15.2 Principled

The probe uses known controls, null baselines, bootstrap intervals, correlation diagnostics, COM alignment, and failure-mode diagnostics. It distinguishes primary readouts from supporting diagnostics.

15.3 Revelatory

The probe does not assume trajectory space will beat COM. It can agree with the COM witness, contradict it, refine it, or fail to produce a usable readout.

16 Strategic Role

Probe T0 helps decide whether Omega is better pursued next as:

1. a quotient/fiber transport object;
2. a trajectory-measure viability object;
3. a viability-kernel erosion object;
4. a recoverability/tube-geometry object;
5. an information-theoretic temporal propagation object;
6. or a structured profile combining several of these.

The current expectation is that the mature theory may use trajectory-space viability and recoverability geometry as a backbone, with quotient/fiber witnesses as finite-state operational projections. Probe T0 treats that as a hypothesis to test rather than an assumption.